

Experiment 2-Random Obstacle: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1985	2.304	1.312	4.923	36.9%	70.4%	82.4%	14.179	28.947	8.588	21.109	0.562	1.307	1.454
Proposed	1977	2.302	0.962	4.139	51.7%	80.2%	87.2%	15.353	28.784	7.460	16.067	0.532	1.286	1.481
PPO-GRU	1990	2.737	1.157	9.105	43.7%	72.8%	82.8%	14.355	28.247	7.342	15.665	0.465	1.175	1.405
MPC	1030	5.882	3.083	15.821	4.3%	19.2%	45.8%	2.876	3.700	1.548	2.756	0.147	0.145	0.187

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1985	99.2%	15	0.8%	0.8%	15	100.0%	0	0
Proposed	2000	1977	98.8%	23	1.1%	1.1%	23	100.0%	0	0
PPO-GRU	2000	1990	99.5%	10	0.5%	0.6%	10	100.0%	0	0
MPC	2000	1030	51.5%	970	48.5%	48.5%	970	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	970	958	98.8%	207	21.6%	2.452
Proposed	970	955	98.5%	145	15.2%	2.518
PPO-GRU	970	961	99.1%	152	15.8%	2.641